Subquadratic sum-of-squares composition formulas over complex and positive-characteristic fields

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Allowed standard prerequisites (editorial).

- Classical Hurwitz-Radon anticommuting orthonormal antisymmetric integer matrices. Author Lemma4, explicitly citing Geramita and Pullman [2] for a self-contained construction; original full bibliography retained.
- Binomial coefficient entropy estimates and Kronecker products. Author proof of Theorem1 and Theorem9; binary entropy is named in the author text.

Author definitions and notation

The problem of Hurwitz [8] asks for which integers n, m, s does there exist a sum-of-squares identity

$$(x_1^2 + \cdots + x_n^2)(y_1^2 + \cdots + y_m^2) = f_1^2 + \cdots + f_s^2, \tag{1}$$

where $f_1, \ldots, f_s$ are bilinear forms in x and y with complex coefficients. Let $\mathbb{F}$ be a field. Define $\sigma_{\mathbb{F}}(n, m)$ as the smallest s such that there exist bilinear $f_1, \ldots, f_s \in \mathbb{F}[x_1, \ldots, x_n, y_1, \ldots, y_m]$ satisfying (1). Furthermore, let $\sigma_{\mathbb{F}}(n) := \sigma_{\mathbb{F}}(n, n)$. Given vectors $u, v \in \mathbb{F}^n$, $\langle u, v\rangle := \sum_{i=1}^n u_i v_i$ is their inner product. For a set S, $\binom{S}{k}$ denotes the set of k-element subsets of S and $\binom{S}{\leq k}$ the set of subsets with at most k elements. $\binom{n}{\leq k} := \sum_{i=0}^k \binom{n}{i}$. $[n]$ is the set $\{1, \ldots, n\}$.

Theorem 1. Let $\mathbb{F}$ be either $\mathbb{C}$ or a field of positive characteristic. Then $\sigma_{\mathbb{F}}(n) \leq O(n^c)$ where $c < 1.62$.

Remark 2(i). If the field has characteristic two, Theorem 1 is trivial. Since $(\sum_i x_i^2)(\sum_j y_j^2) = (\sum_{i,j} x_i y_j)^2$, we have $\sigma_{\mathbb{F}}(n, m) = 1$.

3 Hurwitz-Radon conditions

In this section, we give some well-known properties of σ that we will need later. The definition immediately implies thet $\sigma_{\mathbb{F}}(n,m)$ is symmetric, subadditive, and monotone:

$$\begin{aligned} \sigma_{\mathbb{F}}(n,m) &= \sigma_{\mathbb{F}}(m,n), \\ \sigma_{\mathbb{F}}(n,m_1+m_2) &\le \sigma_{\mathbb{F}}(n,m_1)+\sigma_{\mathbb{F}}(n,m_2), \\ \sigma_{\mathbb{F}}(n,m) &\le \sigma_{\mathbb{F}}(n,m'), \quad m \le m'. \end{aligned} \tag{3}$$

The following lemma gives a characterization of σ in terms of Hurwitz-Radon conditions (4). A proof can be found, e.g., in [13], but we present it for completeness.

Lemma 3. Let $\mathbb{F}$ be a field of characteristic different from two. Then $\sigma_{\mathbb{F}}(n,m)$ equals the smallest s such that there exist matrices $H_1,\dots,H_m \in \mathbb{F}^{n\times s}$ satisfying

$$\begin{aligned} H_i H_i^t &= I_n, \\ H_i H_j^t + H_j H_i^t &= 0, \quad i \ne j, \end{aligned} \tag{4}$$

for every $i,j \in [m]$.

Proof. Let $f_1,\dots,f_s$ be bilinear polynomials in variables $x_1,\dots,x_n$ and $y_1,\dots,y_m$. Then the vector $\bar{f}=(f_1,\dots,f_s)$ can be written as

$$\bar{f} = \sum_{i=1}^{n} \bar{x} H_i y_i,$$

where $\bar{x}=(x_1,\dots,x_n)$ and $H_i \in \mathbb{F}^{n\times s}$. Hence

$$\sum_{k=1}^{s} f_k^2 = \bar{f}\bar{f}^t = \sum_i y_i^2 \bar{x} H_i H_i^t \bar{x}^t + \sum_{i<j} y_i y_j \bar{x}(H_i H_j^t + H_j H_i^t)\bar{x}^t.$$

If the matrices satisfy (4), this equals $\sum_i y_i^2 \bar{x} I_n \bar{x}^t = (y_1^2+\cdots+y_m^2)(x_1^2+\cdots+x_n^2)$, which gives a sum-of-squares identity with s squares. Conversely, if $(y_1^2+\cdots+y_m^2)(x_1^2+\cdots+x_n^2)=\sum f_k^2$, we must have $\bar{x}H_iH_i^t\bar{x}^t = x_1^2+\cdots+x_n^2$ and $\bar{x}(H_iH_j^t+H_jH_i^t)\bar{x}^t=0$. In characteristic different from 2, this is possible only if the conditions (4) are satisfied. □

Given a natural number of the form $n=2^k a$ where a is odd, the Hurwitz-Radon number is defined as

$$\rho(n) = \begin{cases} 2k+1, & \text{if } k = 0 \bmod 4, \\ 2k, & \text{if } k = 1 \bmod 4, \\ 2k, & \text{if } k = 2 \bmod 4, \\ 2k+2, & \text{if } k = 3 \bmod 4. \end{cases}$$

Observe that

$$2\log_2 n \le \rho(n) \le 2\log_2(n) + 2,$$

whenever n is a power of two. Square matrices A_1, A_2 anticommute if $A_1A_2 = -A_2A_1$. A family of square matrices $A_1, \ldots, A_t$ will be called anticommuting if A_i, A_j anticommute for every $i \neq j$. The following lemma is a key ingredient in the proof of Hurwitz-Radon theorem. A self-contained construction can be found in [2].

Lemma 4. For every n, there exists an anticommuting family of $t = \rho(n) - 1$ integer matrices $e_1, \ldots, e_t \in \mathbb{Z}^{n \times n}$ which are orthonormal and antisymmetric (i.e., $e_i e_i^t = I_n$ and $e_i = -e_i^t$).

4 The construction

Let $e_1, \ldots, e_t$ be a set of square matrices. Given $A = \{i_1, \ldots, i_k\} \subseteq [t]$ with $i_1 < \cdots < i_k$, let $e_A := \prod_{j=1}^{k} e_{i_j}$.

Lemma 6. Let $e_1, \ldots, e_t$ be a set of anticommuting matrices. If $A, B \subseteq [t]$ have even size (resp. odd size) then e_A, e_B anticommute assuming $|A \cap B|$ is odd (resp. even).

Proof. Since e_i anticommutes with every e_j, $j \neq i$, but commutes with itself, we obtain

$$e_A e_i = (-1)^{|A \setminus \{i\}|} e_i e_A.$$

This implies that

$$e_A e_B = (-1)^q e_B e_A,$$

where $q = |A| \cdot |B| - |A \cap B|$. Hence if A, B are even (resp. odd) and their intersection is odd (resp. even), q is odd and e_A, e_B anticommute. □

Given integers $0 \leq k \leq t$, a (k,t)-parity representation of dimension s over a field $\mathbb{F}$ is a map $\xi : \binom{[t]}{k} \to \mathbb{F}^s$ such that for every $A, B \in \binom{[t]}{k}$

$$\begin{aligned} &\langle \xi(A), \xi(A) \rangle = 1, \\ &\langle \xi(A), \xi(B) \rangle = 0, \quad \text{if } A \neq B \text{ and } (|A \cap B| = k \bmod 2). \end{aligned} \tag{5}$$

Lemma 7. Let $0 \leq k \leq t$. Over $\mathbb{C}$, there exists a (k,t)-parity representation of dimension $\binom{t}{\leq \lfloor k/2 \rfloor}$. If $\mathbb{F}$ is a field of odd characteristic p, there exists a (k,t)-parity representation of dimension $(p-1)\binom{t}{\leq \lfloor k/2 \rfloor}$. The case of odd characteristic will be proved in the Appendix.

Proof of Lemma 7 over $\mathbb{C}$. Let $0 \leq k \leq t$ be given and $d := \lfloor k/2 \rfloor$. For $a \in \{0,1\}^t$, let $|a|$ be the number of ones in a. Recall that a polynomial is multilinear, if every variable in it has individual degree at most one. We first observe:

Claim 8. There exists a multilinear polynomial $f \in \mathbb{Q}(x_1, \ldots, x_t)$ of degree at most d such that for every $a \in \{0,1\}^t$

$$f(a) = \begin{cases} 1, & \text{if } |a| = k, \\ 0, & \text{if } |a| < k \text{ and } (|a| = k \bmod 2). \end{cases} \tag{6}$$

Proof of Claim. Consider the polynomial

$$g(x_1, \ldots, x_t) := c \prod_{0 \le i < k,\ i = k \bmod 2} \left(\sum_{j=1}^{t} x_j - i \right).$$

Then g has degree d and we can choose $c \in \mathbb{Q}$ so that g satisfies (6). Since we care about inputs from $\{0,1\}^t$, g can be rewritten as a multilinear polynomial f of degree at most d. □

Since f is multilinear, we can write it as

$$f(x_1, \ldots, x_t) = \sum_{C \in \binom{[t]}{\le d}} \alpha_C \prod_{i \in C} x_i,$$

where α_C are rational coefficients. Identifying a subset A of $[t]$ with its characteristic vector in $\{0,1\}^t$, we have

$$f(A) = \sum_{C \subseteq A} \alpha_C.$$

Let $s := \binom{t}{\le d}$. Given $A \in \binom{[t]}{k}$, let $\xi(A) \in \mathbb{C}^s$ be the vector whose coordinates are indexed by subsets $C \in \binom{[t]}{\le d}$ such that

$$\xi(A)_C = \begin{cases} (\alpha_C)^{1/2}, & \text{if } C \subseteq A, \\ 0, & \text{if } C \not\subseteq A. \end{cases}$$

This guarantees

$$\langle \xi(A), \xi(B) \rangle = \sum_C \xi(A)_C \xi(B)_C = \sum_{C \subseteq A \cap B} \alpha_C = f(A \cap B).$$

Hence conditions (6) translate to the desired properties of the map ξ. □

Combining Lemma 6 and 7, we obtain the following bound on σ:

Theorem 9. Let n be a non-negative integer. Let $0 \le k \le \rho(n) - 1$ and $m := \binom{\rho(n)-1}{k}$. Then

$$\sigma_{\mathbb{C}}(n, m) \le n \cdot \binom{\rho(n) - 1}{\le \lfloor k/2 \rfloor}.$$

If $\mathbb{F}$ is a field of odd characteristic p then

$$\sigma_{\mathbb{F}}(n, m) \le (p-1)n \cdot \binom{\rho(n) - 1}{\le \lfloor k/2 \rfloor}.$$

Proof. Let n, k, m be as in the assumption. Let $e_1, \ldots, e_t$ be the matrices from Lemma 4 with $t = \rho(n) - 1$. Let ξ be the (k,t)-parity representation given by the previous lemma. For $A \in \binom{[t]}{k}$, let

$$H_A := e_A \otimes \xi(A),$$

where e_A is defined as in Lemma 6, $\xi(A)$ is viewed as a row vector, and $\otimes$ is the Kronecker (tensor) product. Note that each H_A has dimension $n \times (ns)$ where s is the dimension of the parity representation, and there are $m = \binom{t}{k}$ such matrices H_A. By Lemma 3, it is sufficient to show that the system of matrices H_A, $A \in \binom{[t]}{k}$, satisfies Hurwitz-Radon conditions (4). We have

$$H_A H_B^t = (e_A e_B^t) \otimes (\xi(A)\xi(B)^t) = \langle \xi(A), \xi(B) \rangle \cdot e_A e_B^t.$$

Since every e_i is orthonormal, we have $e_A e_A^t = I_n$. From (5), we have $\langle \xi(A), \xi(A) \rangle = 1$ and hence

$$H_A H_A^t = I_n.$$

If $A \neq B$ then

$$H_A H_B^t + H_B H_A^t = \langle \xi(A), \xi(B) \rangle \cdot (e_A e_B^t + e_B e_A^t). \tag{7}$$

If $|A \cap B| = k \bmod 2$ then $\langle \xi(A), \xi(B) \rangle = 0$ by (5) and hence (7) equals zero. If $|A \cap B| \neq k \bmod 2$ then $e_A e_B^t + e_B e_A^t = 0$. This is because $e_A e_B = -e_B e_A$ by Lemma 6 and that, since e_i are antisymmetric, e_A, e_B are either both symmetric or both antisymmetric. Therefore (7) equals zero for every $A \neq B \in \binom{[t]}{k}$. □

Theorem 1 is an application of Theorem 9.

Proof of Theorem 1. Assume first that n is a power of 16. This gives $\rho(n) = 2\log_2(n) + 1$. Let k be the smallest integer with $n \leq \binom{2\log_2 n}{k} =: m$. From the previous theorem and monotonicity of σ (cf. (3)), we obtain

$$\sigma_{\mathbb{F}}(n) \leq \sigma_{\mathbb{F}}(n, m) \leq cns,$$

where the constant c depends on the field only and $s := \binom{2\log_2 n}{\leq \lfloor k/2 \rfloor}$. We have $k = 2(\alpha + \epsilon_n)\log_2 n$ where $\alpha \in (0, \frac{1}{2})$ is such that $H(\alpha) = 1/2$ (H is the binary entropy function) and $\epsilon_n \to 0$ as n approaches infinity. We also have

$$s \leq 2^{2H(\frac{\alpha+\epsilon_n}{2})\log_2 n} = n^{2H(\frac{\alpha}{2})+\epsilon'_n},$$

where $\epsilon'_n \to 0$. Hence

$$\sigma_{\mathbb{F}}(n) \leq cn^{1+2H(\frac{\alpha}{2})+\epsilon'_n}.$$

The numerical value of α is $0.11\ldots$ which leads to $\sigma_{\mathbb{F}}(n) \leq cn^{1.615+\epsilon'_n} \leq O(n^{1.616})$. If n is not a power of 16, take n' with $n < n' < 16n$ which is. By monotonicity of σ, we have $\sigma_{\mathbb{F}}(n) \leq \sigma_{\mathbb{F}}(n')$. □

A Proof of Lemma 7 in positive characteristic

Given non-negative integers $\bar{n} = (n_1, \ldots, n_d)$ let $B(\bar{n})$ be the $d \times d$ matrix $\{B(\bar{n})_{i,j}\}_{i,j \in [d]}$ with

$$B(\bar{n})_{i,j} = \binom{n_j}{i-1}.$$

We assume that $\binom{n}{k} = 0$ whenever $n < k$; this guarantees $\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!}$.

Lemma 16. If $\bar{n} = (r, r+2, \ldots, r+2(d-1))$ for some non-negative integer r then $\det(B(\bar{n})) = 2^{\binom{d}{2}}$.

Proof. We claim that

$$\det(B(\bar{n})) = \left(\prod_{i=1}^{d-1} i!\right)^{-1} \det(V(\bar{n})),$$

where $V(\bar{n})$ is the Vandermonde matrix with entries $V(\bar{n})_{i,j} = n_j^{i-1}$. To see this, multiply every i-th row of $B(\bar{n})$ by $(i-1)!$ to obtain matrix $B'(\bar{n})$ with

$$\det(B'(\bar{n})) = \left(\prod_{i=1}^{d-1} i!\right) \det(B(\bar{n})).$$

An i-th row r_i of $B'(\bar{n})$ is of the form $(n_1^{i-1} + g_i(n_1), \ldots, n_d^i + g_i(n_d))$ where g_i is a polynomial of degree $< (i-1)$. This means that r_i equals the i-th row of $V(\bar{n})$ plus a suitable linear of combination of the preceding rows of $V(\bar{n})$. Therefore, $\det(B'(\bar{n})) = \det(V(\bar{n}))$. Given $\bar{n}$ as in the assumption, we obtain

$$\begin{aligned}\det(V(\bar{n})) &= \prod_{1 \le j_1 < j_2 \le d} (n_{j_2} - n_{j_1}) = \prod_{1 \le j_1 < j_2 \le d} (2j_2 - 2j_1) \\ &= 2^{\binom{d}{2}} \prod_{1 \le j_1 < j_2 \le d} (j_2 - j_1) = 2^{\binom{d}{2}} \prod_{i=1}^{d-1} i!.\end{aligned}$$

This shows that $\det(B(\bar{n})) = 2^{\binom{d}{2}}$. □

Lemma 17. Let p be an odd prime. Given $0 \le k \le t$, there exists a multilinear polynomial $f \in \mathbb{F}_p(x_1, \ldots, x_t)$ of degree at most $d = \lfloor k/2 \rfloor$ such that for every $a \in \{0,1\}^t$

$$f(a) = \begin{cases} 1, & \text{if } |a| = k, \\ 0, & \text{if } |a| < k \text{ and } (|a| = k \bmod 2). \end{cases}$$

Proof. We look for f of the form $f = \sum_{j=0}^{d} c_j S_t^j$ where S_t^j is the elementary symmetric polynomial $S_t^j = \sum_{|A|=j} \prod_{i \in A} x_i$. Given $a \in \{0,1\}^t$,

$$f(a) = \sum_{j=0}^{d} c_j \binom{|a|}{j} \bmod p.$$

We are therefore looking for a solution of the linear system

$$B(\bar{n})^t (c_0 \ldots, c_d)^t = (0, \ldots, 0, 1)^t,$$

where $\bar{n} = (0, 2, \ldots, 2d)$, if k is even, and $\bar{n} = (1, 3, \ldots, 2d+1)$, if k is odd. By the previous lemma, $B(\bar{n})$ is invertible over $\mathbb{F}_p$ and such a solution exists. □

Lemma 18. If $\mathbb{F}$ is a field of odd characteristic p, there exists a (k,t)-parity representation of dimension $(p-1)\binom{t}{\leq \lfloor k/2 \rfloor}$.

Proof. If every element of $\mathbb{F}_p$ has a square root in $\mathbb{F}$, the proof is the same as over $\mathbb{C}$. In general, proceed as follows. Since every non-zero element of $\mathbb{F}_p$ is a sum of at most $(p-1)$ ones, we can write

$$f(x_1, \ldots, x_t) = \sum_{C \in \mathcal{C}} \prod_{i \in C} x_i,$$

where $\mathcal{C}$ is a multiset of $s \leq (p-1)\binom{t}{\leq d}$ subsets of $[t]$. For $A \in \binom{[t]}{k}$, let $\xi(A) \in \mathbb{F}^s$ be a vector whose coordinates are indexed by elements C of $\mathcal{C}$ so that

$$\xi(A)_C = \begin{cases} 1, & \text{if } C \subseteq A, \\ 0, & \text{if } C \not\subseteq A. \end{cases}$$

□

References

[1] Marco L. Carmosino, Russell Impagliazzo, Shachar Lovett, and Ivan Mihajlin. Hardness amplification for non-commutative arithmetic circuits. In Proceedings of the 33rd Computational Complexity Conference, CCC '18, 2018.

[2] A. Geramita and N. Pullman. A theorem of Hurwitz and Radon and orthogonal projective modules. Proceedings of The American Mathematical Society, 42:51–51, January 1974. doi: 10.1090/S0002-9939-1974-0332764-4.

[3] I. Haviv. On minrank and the Lovász Theta function. In Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques, 2018.

[4] I. Haviv. Topological bounds on the dimension of orthogonal representations of graphs. European Journal of Combinatorics, 81:84–97, 2019.

[5] P. Hrubeš. On families of anticommuting matrices. Linear Algebra and Applications, 493:494– 507, 2016.

[6] P. Hrubeš, A. Wigderson, and A. Yehudayoff. Non-commutative circuits and the sum of squares problem. In STOC' 10 Proceedings of the 42nd symposium on Theory of Computing, pages 667–676, 2010.

[7] P. Hrubeš, A. Wigderson, and A. Yehudayoff. An asymptotic bound on the composition number of integer sums of squares formulas. Canadian Mathematical Bulletin, 56:70–79, 2013.

[8] A. Hurwitz. Über die Komposition der quadratischen Formen von beliebigvielen Variabeln. Nach. Ges. der Wiss. Göttingen, pages 309–316, 1898.

[9] L. Lovász. On the Shannon capacity of a graph. IEEE Trans. Inform. Theory, 25(1):1–7, 1979.

[10] N. Nisan. Lower bounds for non-commutative computation. In Proceeding of the 23th STOC, pages 410–418, 1991.

[11] J. Radon. Lineare scharen orthogonalen Matrizen. Abh. Math. Sem. Univ. Hamburg, 1(2-14), 1922.

[12] D. B. Shapiro. Quadratic forms and similarities. Bull. Amer. Math. Soc., 81(6), 1975.

[13] D. B. Shapiro. Compositions of quadratic forms. De Gruyter expositions in mathematics 33, 2000.